

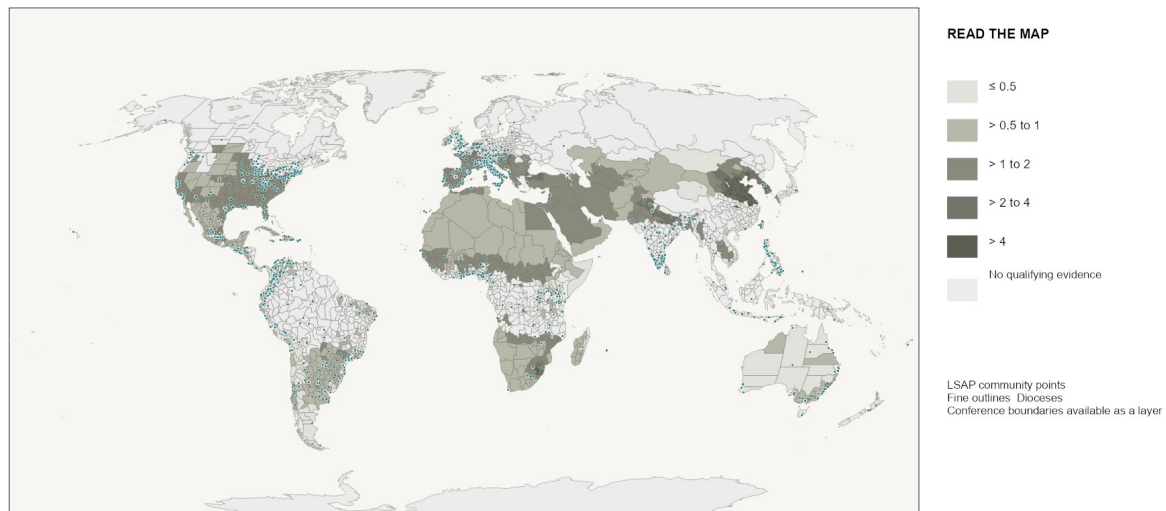
Annual nitrogen dioxide across dioceses

This report explains the source data, executed workflow and interpretation of annual nitrogen dioxide across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Glissenaar et al., 2024, 2025; KNMI, n.d.).

Map P02 | Documentation date 6 October 2026 | Units 10^{15} molecules/cm² | Air year 2021

Annual nitrogen dioxide across dioceses

Area weighted tropospheric NO₂ - 2021 - 10^{15} molecules/cm²



Shading requires 50% valid polygon coverage. Monthly means weighted by calendar days. At least 9 qualifying months and all quarters. Column density differs from ground-level exposure.

KNMI ESA CCI+ TROPOMI L3 - v1.26 - 0.2° grid

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P02

Figure P02 World overview exported from the completed ArcGIS Pro layout.

Shading requires 50% valid polygon coverage. Monthly means weighted by calendar days. At least 9 qualifying months and all quarters. Column density differs from ground-level exposure.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Glissenaar et al., 2024, 2025; KNMI, n.d.).

1. Download all twelve 2021 KNMI ESA CCI+ TROPOMI L3 v1.26 monthly files (fv1.0 filename version), native 900 × 1800 grid.
2. Select monthly cells with qa_L3=1, land/transition mask, finite column value and fractional effective monthly coverage × calendar days ≥10. This effective coverage is not a literal count of valid days.
3. Calculate calendar-day-weighted means of qualifying monthly columns. Require at least nine qualifying months and representation from all four quarters. Preserve negative retrieval values; divide by 10¹⁵ to express molecules/cm² in readable units.
4. Export annual concentration and qualifying-month-count rasters; audit both CRS and grids. Aggregate annual concentration using geodesic cell areas and fractional overlaps. Retain weighted spatial median and full-polygon valid coverage.
5. Classify diocesan AirDisplay at 0.5, 1, 2 and 4 ×10¹⁵ molecules/cm². Overlay copied LSAP dots, diocesan outlines and optional conference boundaries.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\Sigma(c_i \times a_i) / \Sigma(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO₂ implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
AirMean	Area weighted mean over valid overlap
AirMedian	Area weighted spatial median
ValidAreaPct	Valid overlap as percent of full diocesan polygon
ValidCells	Number of contributing native raster cells
AirDisplay	Mean where coverage is at least 50 percent; otherwise null
AirStory	Interpretation narrative and coverage
AirYear and AirUnits	2021 and concentration units

Analysis dataset: PollutionAnalysis.gdb/Dioceses_NO2. Styles/P02.lyrx and Styles/P02.mapx preserve layer and map definitions. Map_Exports/P02_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Retained raster audit reports 1,404 dioceses meeting the 50 percent shading threshold. These counts differ from the number with any computed statistics (LSAP Pollution Atlas, 2026, raster_audit.json).

Interpretation and data handling

A cell with nine qualifying months is an observed-month estimate, not complete annual sampling. Tropospheric column density is different from surface concentration or personal exposure. KNMI source.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

Glissenaar, I. A., Anglou, I., Boersma, K. F., & Eskes, H. (2024). ESA CCI+ TROPOMI L3 monthly mean NO2 columns [Data set]. Royal Netherlands Meteorological Institute. Version 1.26, filenames fv1.0; 2021 monthly files. [Source link](#)

Glissenaar, I., Boersma, K. F., Anglou, I., Rijdsdijk, P., Verhoelst, T., Compernelle, S., Pinardi, G., Lambert, J.-C., Van Roozendaal, M., & Eskes, H. (2025). TROPOMI Level 3 tropospheric NO2 dataset with advanced uncertainty analysis from the ESA CCI+ ECV precursor project. *Earth System Science Data*, 17, 4627–4650. [Source link](#)

KNMI. (n.d.). ESA CCI+ NO2 TROPOMI level 3 data. Download and version documentation. Accessed 6 October 2026. [Source link](#)

Esri. (n.d.). Calculate Geometry Attributes. ArcGIS Pro documentation. Reference for geodesic area measurement. Accessed 6 October 2026. [Source link](#)

LSAP Pollution Atlas. (2026). Internal source copies, executed analysis scripts and validation records [Project records]. LSAP_Pollution.aprx; FoundationCopies.gdb; PollutionAnalysis.gdb; foundation_manifest.json; raster_audit.json; freshwater_aggregate_QA.json; waste_QA.json; cleanup_audit.json; project_QA.json. Project root: Atlas Analyses/Pollution. Internal provenance is distinct from published source citations.